

Submersions, Immersions and Embeddings

1.0.1 Local Diffeomorphism

Definition 1.0.1

If M and N are smooth manifolds with or without boundary, a map $F : M \rightarrow N$ is called a **local diffeomorphism** if every point $p \in M$ has a neighborhood U such that $F(U)$ is open in N and $F|_U : U \rightarrow F(U)$ is a diffeomorphism.

Theorem 1.0.2 ▶ Inverse Function Theorem for Manifolds

Suppose M and N are smooth manifolds, and $F : M \rightarrow N$ is a smooth map. If $p \in M$ is a point such that dF_p is invertible, then there are connected neighborhoods U_0 of p and V_0 of $F(p)$ such that $F|_{U_0} : U_0 \rightarrow V_0$ is a diffeomorphism.

Note.

也就是说, 光滑映射在微分映射可逆的点附近是微分同胚.

1.1 Submersions

Definition 1.1.1

- If $\pi : M \rightarrow N$ is any continuous map, a **section of π** is a continuous right inverse for π , i.e., a continuous map $\sigma : N \rightarrow M$ such that $\pi \circ \sigma = \text{Id}_N$:

$$\begin{array}{c} M \\ \pi \downarrow \uparrow \sigma \\ N \end{array}$$

- A **local section of π** is a continuous map $\sigma : U \rightarrow M$ defined on

some open subset $U \subseteq N$ and satisfying the analogous relation $\pi \circ \sigma = \text{Id}_U$.

Theorem 1.1.2 ▶ Local Section Theorem

Suppose M and N are smooth manifolds and $\pi : M \rightarrow N$ is a smooth map. Then π is a smooth submersion if and only if every point of M is in the image of a smooth local section of π .

Proof sketch.

光滑淹没的标准坐标表示 $\pi(x^1, \dots, x^n, x^{n+1}, \dots, x^m) = (x^1, \dots, x^n)$ 下, 将坐标邻域缩小成方块 $C_\varepsilon = \{x : |x^i| < \varepsilon, i = 1, \dots, m\}$, 那么坐标像也是方块 $C'_\varepsilon = \{y : |y^i| < \varepsilon, i = 1, \dots, n\}$. 通过 C'_ε 上的点末尾坐标补零 $\sigma(x^1, \dots, x^n) = (x^1, \dots, x^n, 0, \dots, 0)$, 可以方便地给出一个截面. 反过来, 若 p 点位于某个截面像, 则 $\pi \circ \sigma = \text{Id}_U$ 求导 $d\pi_p \circ d\sigma_q = \text{Id}_{T_q N}$ 立即给出 π 在 p 点是淹没.

Proposition 1.1.3 ▶ Properties of Smooth Submersions

Let M and N be smooth manifolds, and suppose $\pi : M \rightarrow N$ is a smooth submersion. Then π is an open map, and if it is surjective it is a quotient map.

Proof sketch.

截面性质 $\pi \circ \sigma = \text{Id}_U$ 给出 $\pi \circ \sigma(\sigma^{-1}(W)) \subseteq \pi(W)$. 截面的连续性给出 $\sigma^{-1}(W)$ 是开集, 若 W 是开集. 由此给出像点的开邻域, 说明淹没是开的. 连续而开的映射是商映射.

Theorem 1.1.4 ▶ Characteristic Property of Surjective Smooth Submersions

Suppose M and N are smooth manifolds, and $\pi : M \rightarrow N$ is a surjective smooth submersion. For any smooth manifold P with or without boundary, a map $F : N \rightarrow P$ is smooth if and only if $F \circ \pi$ is smooth:

$$\begin{array}{ccc} M & & \\ \pi \downarrow & \searrow F \circ \pi & \\ N & \xrightarrow{F} & P. \end{array}$$

Proof sketch.

局部截面 σ 的存在性, 允许我们局部地将 F 提升到 M 上 $F \circ \pi$, 使得它可以被 σ 还原: $F|_U = (F \circ \pi) \circ \sigma$.

Theorem 1.1.5 ▶ Passing Smoothly to the Quotient

Suppose M and N are smooth manifolds and $\pi : M \rightarrow N$ is a surjective smooth submersion. If P is a smooth manifold with or without boundary and $F : M \rightarrow P$ is a smooth map that is constant on the fibers of π , then there exists a unique smooth map $\tilde{F} : N \rightarrow P$ such that $\tilde{F} \circ \pi = F$:

$$\begin{array}{ccc} M & & \\ \pi \downarrow & \searrow F & \\ N & \xrightarrow{\tilde{F}} & P. \end{array}$$

Proof sketch.

满的光滑淹没是商映射, 商的泛性质刻画了唯一的满足交换图的连续映射 $\tilde{F}$, 再通过满的光滑淹没刻画其连续性.